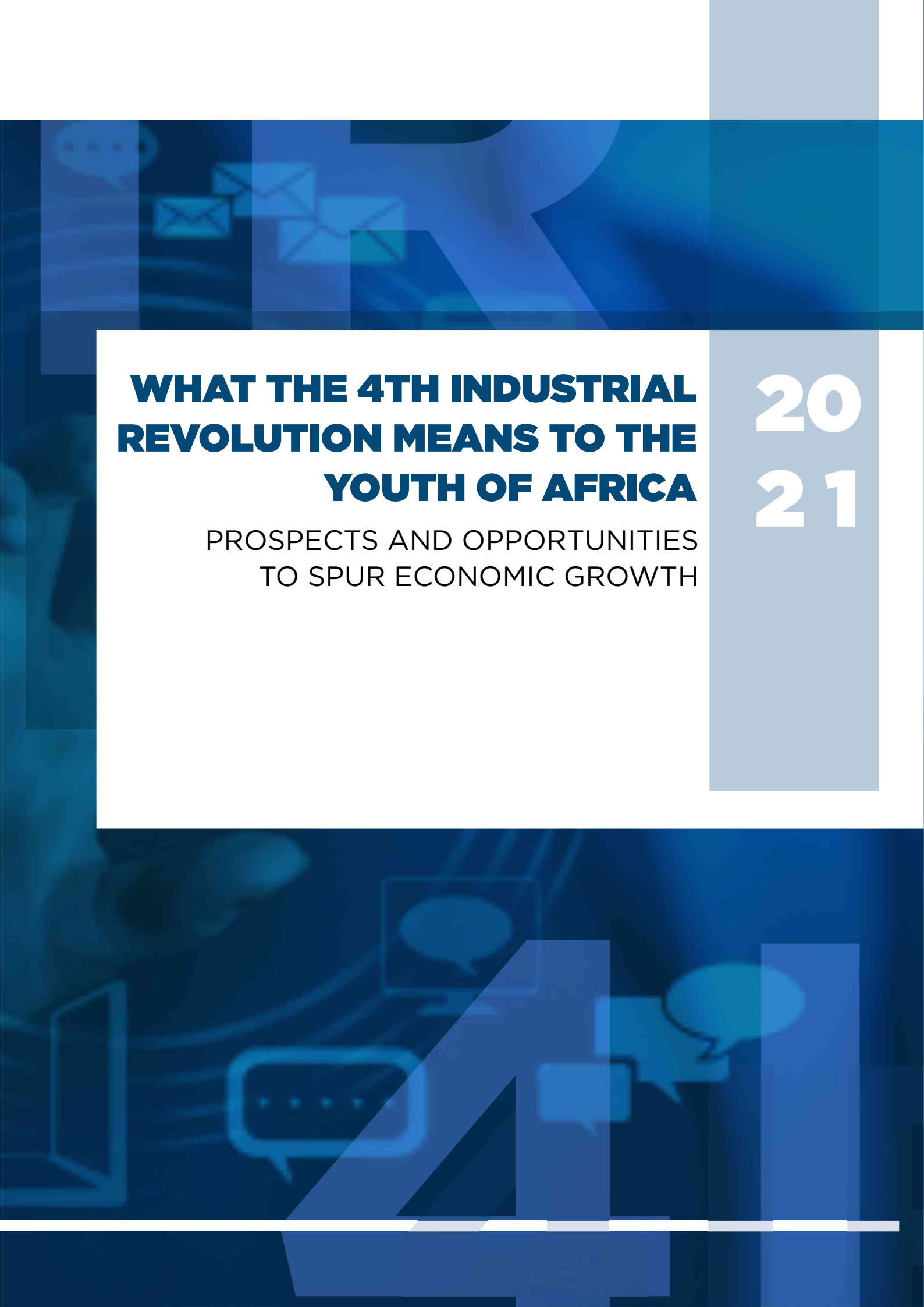




WHAT THE 4TH INDUSTRIAL REVOLUTION MEANS TO THE YOUTH OF AFRICA

PROSPECTS AND OPPORTUNITIES
TO SPUR ECONOMIC GROWTH

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ABSTRACT

Among the most distinguishing characteristics of our generation is the rapid growth of technological capabilities. Our social and political relationships have been and will continue to be influenced by the digital revolution, which has impacted employment and labour, how individuals live, and how they connect with others. Our method of working is evolving as a result of the Fourth Industrial Revolution (4IR). European and Western leaders are wrestling with this prospect, but African and developing nations, in general, appear to be facing an even more bleak future than they are. Technology advancements will reduce the length of the conventional road to economic transformation through manufacturing. A large number of jobs will be eliminated. It is a question of when, not if, something will happen.

On the other hand, technology will alter Africa and open up new ways for bypassing the old structures. In order to do this, the international communities, African countries, and the technological community must work together to harness the potential of the Fourth Industrial Revolution (4IR). The inevitable reality is that if this does not occur soon, a new technical disparity will emerge, which would increase the gap between the developed and developing worlds even more (African Center for Economic Transformation, 2020). What does the Fourth Industrial Revolution mean to Africa's youth? What are the chances and opportunities it presents for promoting African economic growth? In order to offer answers to these concerns, the study analyses the potential that 4IR can bring for economic growth paths and the opportunities that the new digital technologies will provide for Africa's economy and its future generations.

Job opportunities in Africa are relevant right now, and they will remain so in the future, considering the number of young Africans projected to enter the labour force. Unfortunately, due to the Covid-19 global pandemic, labour income has dropped by an estimated 10% (World Bank, 2020). According to the International Labour Organization (2020), youth aged 15 to 24 entering the labour force are especially most affected due to an absence of economic opportunities. According to the ILO report, youth unemployment has increased significantly, particularly in Southern and Northern Africa. According to the ILO, in 2018, one-third of Africa's employed population lived in extreme poverty (ILO, 2019). Prior to the outbreak of the Covid-19, Sub-Saharan African economies were expanding rapidly and providing wage and salaried jobs to a sizable portion of their labour force (Fox & Gandhi, 2021). Though official salary and wage jobs were preferable because they provided job security, most of Africa's labour force continues to work in the informal sector. Better jobs are available in developed countries, and countries grow richer by developing productive jobs, in contrast to Africa, where most jobs are low-productivity activities (Fox & Signé, 2021).

The economic challenge post-Covid-19 is to rebuild the economy, including expanding employment opportunities for all Africans. The African youth face unique challenges in entering the labour force and finding work. This necessitates resuming and intensifying inclusive economic transformation in Africa. However, increasing automation and the emergence of digital technologies pose a serious challenge to developing countries' traditional economic routes (BBC, 2019). As artificial intelligence (AI), 3D printing, advanced robotics and other contemporary technologies outmanoeuvre and outbid labour, developing countries' ability to deploy great, less expensive work to their profitable additional benefit reduces. African economies are already plagued by high unemployment, underemployment, and poverty. On a continent where the population is predicted to increase, African governments face the looming problem of employing a large, unskilled workforce in more productive activities (ILO, 2016).

According to the Economist (2020), the population of Africa is expected to double by 2050, from 1.26 billion people in 2017 to 2.53 billion people in 2050. The Ibrahim Forum also published a report in 2019 stating that by 2100, Africa's population will triple to 4.47 billion people, with a substantial youth bulge in the population (Ibrahim Forum Report, 2019). Africa now has a population of 17% of the world's total; by 2050, it will have 26%, and it will be 40% by 2100. By 2050, the population of Africa will be 3.5 times that of Europe and 6.8 times that of Europe by 2100.

(United Nations, 2017). By 2050, the population of Nigeria alone will be more than half that of Europe, and 1.2 times that by 2100 (United Nations, 2017).

Because of the extent and pace of invention, it is commonly expected that the cyber-physical technologies of the next generation would dramatically affect production capacity, trade, consumption, living standards, and the pricing of products and services - much quicker than past industrial revolutions. The 4IR technological tsunami has already begun to wash up on the shores of Africa, having been developed in wealthy nations where adoption is quick (Fox & Signé, 2021). Though Africa has not digitally transformed as quickly as other developing and developed regions, significant advances in information and communications technology have occurred (ICT). Mobile technologies and services create millions of jobs and add substantial economic value to the region, demonstrating that digitization aids in increasing efficiency and growth (International Telecommunication Union, 2021). Many 4IR technologies will provide Africa with a platform for innovation globally, and African leaders must act quickly to develop policies to capitalize on the opportunities these technologies present.

Africa's fast-expanding population may be an excellent tool for socioeconomic development if quality education, skilling, and employment policies are implemented. However, to reap the benefits of this potential demographic dividend, African countries must quickly seize new possibilities and overcome obstacles. Jobs in all industries will be affected by this transformation but at various phases and variable degrees. Africa will not become digitized immediately, but the global effect of 4IR makes the economic change Africa needs to exploit its advantages and achieve its potential much more urgent.

WHAT IS THE FOURTH INDUSTRIAL REVOLUTION?

WHAT THE 4TH INDUSTRIAL
REVOLUTION MEANS TO THE
YOUTH OF AFRICA

The term "Fourth Industrial Revolution" is reported to have been coined by Klaus Schwab of the World Economic Forum. According to Schwab, the 4IR is the emergence of technologies that combine the physical, digital and biological, disrupting all countries' sectors. All goods and services can benefit from these technologies in terms of efficiency, speed and long-term viability. The 4IR denotes a combination of advancements in Artificial intelligence, automation, 3D printing, genetic engineering, quantum computing, the Internet of Things, and other technologies (Schwab, 2016).

Schwab (2016) elaborates that the Fourth Industrial Revolution has three different characteristics: velocity, scope, and system effect. Velocity is the rate at which the 4IR technologies grow and advance. While the steam engine took decades, nearly a century, to completely transform the global industry in the 18th century, there is about a century between 1IR, 2IR, and 3IR, 4IR advances; new advancements are coming onto the market less than 50 years after 3IR began. They are spreading at a breakneck speed as a result of globalization. The term "scope" refers to how these technologies affect a wide range of businesses, sectors, and employment. For example, additive manufacturing, commonly known as 3D printing, may be used to make various light and heavy objects, including circuit boards, wind turbines, customized sports shoes, and everything in between. Finally, systems effect is the magnitude and complexity of the changes currently taking place and are projected to continue throughout the whole production, management, and governance system. Many patterns of human existence, particularly African economies, are predicted to be disrupted by this combination of velocity, scope, and impact (Schwab, 2016).

THE FOURTH INDUSTRIAL REVOLUTION AND THE WORLD OF WORK

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Globally, fundamental factors influence firms' decision-making processes on the adoption of the Fourth Industrial Revolution. The cost of labour in relation to capital is an essential factor to consider when deciding whether or not to automate. Additionally, market innovation, the sort of work to be automated, and the kind of capital investment connected with the automation opportunity are all aspects to consider (Karr, Loh & San Andres, 2020).

Even before the Covid-19 pandemic, the consensus was that the world economy had reached a new era of automation known as the Fourth Industrial Revolution, which was characterized by breakthroughs in artificial intelligence, the Internet of Things, software robotics and autonomous hardware, and advanced data systems that enabled predictive and real-time analytics (Kudyba, 2020). Businesses in Africa and the rest of the world are now facing an uncertain future in which they must adjust their operations and business strategies to the "New Normal" conditions. As the COVID-19 issue expanded throughout the world, its impact on societies, businesses, and employees produced new factors that might encourage or hinder the shift to 4IR-enabled automation. The probable contributors include firm-level organizational changes, constrained labour supply and efficiency, and second-order effects of government actions addressing the COVID-19 crisis (Fox & Signé, 2021).

Ground-breaking technology breakthroughs are affecting nearly every industry globally, and the scope of these changes is transforming whole production, management, and governance systems. While the potential for entire new sectors, employment positions, and expertise is enormous, these technologies also provide a rising threat of redundancies within these domains, as current methods are rendered irrelevant by breakthrough technology. Due to rapid technological progress, it is difficult to predict how the 4IR will affect organizations and their personnel. Many sectors see the arrival of new technologies and capabilities that open up new avenues for meeting current demands (Sutherland, 2020).

In its report, the McKinsey Global Institute (2017) estimated that the Fourth Industrial Revolution

would displace 800 million employees in 42 nations, or one-third of the workforce. According to the McKinsey report, many low-skilled, repetitive tasks are being automated, with the trend beginning in high-wage countries and fast spreading to developing countries. Notably, the report shows that at least one-third of the constituent tasks in about 60% of jobs may be automated, indicating significant workplace reforms and adjustments for all workers (McKinsey Global Institute, 2017). We glean from this report that automation will have a fundamental impact on how functions at the workplace are carried out and by who or what. Admittedly, artificial intelligence and automation will increase productivity and economic growth; however, millions of individuals may be forced to change occupations or upgrade their abilities.

It is noteworthy that even when specific jobs are mechanized, employment in certain occupations may not decline but may instead increase as people take on new responsibilities. Because robots can now match human performance, automation will have a reduced impact on professions that require managing people, using expertise, and maintaining social relationships (McKinsey Global Institute, 2017). Existing employees would have to upskill where need be to keep their jobs. Youths must obtain employable skills that are relevant for the new workplace to secure a job successfully.

Cirera (2016) indicates that insufficient education can impair a firm's ability to transfer knowledge into innovation, implying that a skilled staff is essential for capitalizing on the fourth Industrial Revolution. (Ayentimi & Burgess, 2019).

The 4IR can play a significant role in promoting Africa's development and hence employment creation. What matters is whether or not the countries have the necessary capabilities to lead this transition. A recent report by the World Economic Forum (2017) asserts that in addition to STEM skills, which are considered to be the foundation of the 4IR skills, other essential competencies include complex problem-solving abilities, entrepreneurship and business skills, critical thinking and active listening skills, cognitive skills (logical reasoning and creativity), social skills and system skills (such as judgment).

Formerly, manufacturing's exportability and formality were somewhat unique: exported manufactured products provided foreign cash and taxes, which were reinvested into more technologically advanced capital and foreign technical expertise, further enhancing productivity. On the other hand, domestic services, marketplaces, and driving consumed much urban labour in Africa and remained relatively informal and non-exportable, providing little chances to boost a country's productivity (Ayentimi & Burgess, 2019). However, with the advent of the 4IR technologies, digital platforms and the internet, many services that add value can now be exported and formalized.

Conventional specialized services like accountancy, insurance, financial assistance, and other commercial services may now be exported more easily via the internet and paid for using digital currencies (Abe, Abe & Adisa, 2021). On the other hand, the development of the sharing economy is providing the most exciting prospects for structural transformation via services exports. The introduction of mobile money and digital platforms as the go-to-market channel has resulted in the formalization of services in emerging economies. We see formalization in domestic industries such as transportation (like Uber) and export services through electronic marketing platforms like eBay and Alibaba.

Even though African nations made progress in employment change and economically, young people continue to encounter difficulties in finding work (Pityana, 2019). These are both economic and social issues. The concern seems to be whether 4IR technologies will reverse the current trend and how and who will benefit.

The world of work is changing because of the rapid development in internet connectivity and automation. The rising use of robots in industrial activities, notably in the manufacturing sector, has put many jobs at risk. In emerging countries, an estimated 66% of all jobs are in jeopardy (Frey, Osborne & Holmes, 2016). Current jobs in African nations such as Angola and Ethiopia, which account for roughly 50% and 44% of total employment, are vulnerable to automation (Randow, 2016). As a result, Randow (2016) argues that by shifting people into manufacturing jobs, African countries will lose their cost advantage and, potentially, their ability to achieve rapid economic growth. This is due to the emergence of disruptive technologies that are creating changes in the company and work paradigms—for example, the rise of sharing economy companies like Uber impacts established industries like the taxi sector.

According to Brown (2020), Africa's young people are a valuable resource as well as a ticking time bomb. By 2025, almost half of these energetic young individuals will be jobless, discouraged, or economically inactive. A well-publicized portion of the population is moving in search of work, frequently via perilous routes. Unfortunately, 40% of young people joining rebel and terror organizations are motivated by a lack of economic opportunities (Brown, 2020).

The anxiety of the implications of the 4IR has grown in many emerging nations in Sub-Saharan Africa because of the influence it would have on job displacement, skills discontinuities, and present rising unemployment and skill scarcities, which are already subverting economic transformation and long-term progress (World Economic Forum Africa, 2016).

THE 4IR TO SPUR ECONOMIC GROWTH IN AFRICA

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The first industrial revolution (1IR) saw the use of machines to replace manual labour, whereas the second industrial revolution (2IR) saw the expansion of manufacturing, transportation and communication technology (Kim, 2018). The primary engine of the third industrial revolution (3IR) was digitalization and development in electronics and technology (Naudé 2017). However, the 4IR is propelled by advances in artificial intelligence, artificial intelligence robotics, genetics, nanotechnology, 3D printing, biotechnology, and fuelled by expanding many capacities and technologies (World Economic Forum, 2017). This technical progress is claimed to be the primary driver of social and economic progress, and it is expected to drive the global economic and social revolution. The 4IR is said to encourage the development of general-purpose technology to boost productivity in various industries and businesses (World Economic Forum 2017).

The 4IR aims to increase per capita income while raising living standards, productivity, and consumption (Makridakis, 2017). The 4IR will impact all sectors and industries' social structures and work systems (Naudé, 2017). In most established or advanced countries, technological innovations and their application have been the driving engine of productivity growth, resulting in essential breakthroughs and economic and social development (World Economic Forum, 2017). However, throughout previous industrial revolutions, technological change and its implementation in Sub-Saharan Africa have been uneven, resulting in little social and economic transformation (The Africa Competitiveness Report, 2017).

One of the distinguishing features of emerging economies in Sub-Saharan Africa that differs from developed economies is the existence and domination of the informal sector, which is underpinned mainly by small and micro-enterprises such as small-holder farming, petty-trading, handwork, food-sellers, and small agro-processing businesses (Osei-Boateng & Ampratwum, 2011). According to the International Labour Organization, the informal sector in Sub-Saharan Africa accounts for around ierra Leone, Ghana, Zimbabwe and other African countries (ILO, 2016).

72% to 80% of overall employment and three-quarters of non-agricultural work in the region (Potts 2008). The informal industry in the region extends from less than 30% of business activities in South Africa to 60% in Sub-Saharan Africa, the informal sector has so been identified as a possible source of employment. The effect of the 4IR on the informal sector, which is the principal source of jobs for young people in the region, is a crucial reason for concern for emerging economies across Africa (ILO, 2016).

According to Osei-Boateng and Ampratwum (2011), the unregulated nature of the informal sector has led to limited government revenue mobilization efforts across the region. Information and Communication Technology provides the avenue to reorganize and improve the manufacturing scale of small, micro and agro-processing companies and the formation of new businesses. Sustained use of ICT in the informal economy will contribute to its collapse by bringing more individuals into the formal labour market (Naude, 2017).

Based on the preceding, the 4IR's importance to Sub-Saharan Africa cannot be overstated in light of the prospects it presents. The 4IR can influence all businesses and countries, regardless of their location or development stage. The 4IR offers limitless potential for Africa's economic restructuring, particularly in the informal sector. The 4IR offers the informal economy a significant chance to use simple technologies to boost productivity and help formalize all industries. For example, a small company could be connected to supply chains and have access to marketing data such as the price of commodities, demand and supply. According to Naudé (2017), given the untapped market in Sub-Saharan Africa, the 4IR could position Sub-Saharan Africa, and Africa in general, to pursue their failed industrialization agenda based on digital services exports, while also supporting the promotion of new business models across the continent for intra-regional trade.

Africa has excellent possibilities to solve its human-capital issues in the health industry with 4IR technologies. Low health-related human-development results will stifle productivity at both the

individual and national levels, and 4IR technologies can assist solve these issues. Africa may tackle some of these problems by leapfrogging conventional development obstacles such as a lack of physicians or basic infrastructure by applying 4IR technologies. Artificial Intelligence may be used to address various health issues in Africa, such as a shortage of physicians and inadequate infrastructure, such as unstable electricity that fails to power medical equipment.

HOW AFRICAN COUNTRIES ARE USING 4IR TO SOLVE EMPLOYMENT PROBLEMS

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In recognition of the significance of entrepreneurship for economic development, jobs, and poverty reduction, respective countries have begun implementing supporting public policies to streamline business rules for small enterprises and startups. African countries leverage venture capital arrangements to set up startup hubs to provide funding for accelerating early-stage startups, empowering entrepreneurs, and encouraging innovations that facilitate economic development, poverty reduction, and job creation (Fal, 2013).

The venture capital landscape has become a recognized investment mechanism to attract foreign investment into the continent, especially the fintech industry. According to a 2020 report by the African Private Equity and Venture Capital Association, Africa's venture capital landscape has evolved over the last two to three decades. Most African countries set up startup hubs and secure venture capital arrangements to fund innovative ideas.

In 2018, Tunisia became the first African country to pass a Start-up Act to support innovative ideas (International Network for Small and Medium-Sized Enterprises, 2021). The startup act was a part of the more comprehensive strategy called Digital Tunisia 2020. More than 280 Startups took advantage of the Startup Act. In 2019, Senegal became the second African country to pass a Start-up Act (Disrupt Africa, 2019). The Senegal Startup Act aims to promote innovation that can have a ripple effect on the national economy under the Digital Senegal 2025 strategy (Techbuild Africa, 2019). The Senegalese government set up the Délégation de l'Entrepreneuriat Rapide ([DER](#)) to help innovative local tech startups overcome constraints such as the lack of early solid support structures accelerators and access to early-stage funding. The government invested \$2million to support 40 tech startups (Disrupt Africa, 2019). The tech businesses that benefitted included Eyone, Firefly, Shopmeaway, Bayseddo, Awalebiz, Deedo, mTick, Comparez, PayExpresse and Afrikamart.

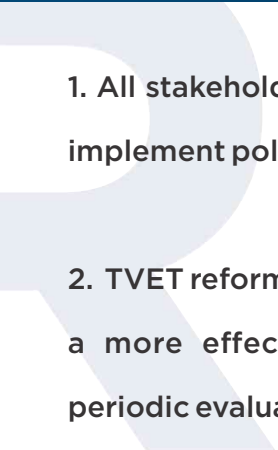
In Kenya, the tech ecosystem, popularly known as Silicon Savannah, has made the Capital Nairobi a favourite spot for venture capital investors (Fififinance, 2020). The tremendous economic growth in

Kenya is partly due to the innovations created in 2010 through iHub. Ever since then, in Kenya's major cities, tech hubs have spread at a significant rate. Some of the technology hubs are Swahilibox in Mombasa, Kisumu's Lakehub, Eldoret's Dlab Hub, Voi's Sote Hub, Machakos' Ubunifu and the Nyeri Mt. Kenya Hub. The Kenyan government has funded tech startups intending to alleviate poverty and unemployment through the Youth Enterprise Development Fund. For instance, the fund has provided support to its fintech company, Cellulant, to provide Pan-African digital payment services to people from different sectors. Some other tech startups that have been supported include CarePay (providers of innovative insurtech solutions), M-Kopa (providing clean solar energy), Copia Global (providing mobile commerce service), Twiga Foods (a platform that connects farmers to potential buyers), and Sendy (connecting businesses and drivers of different types of vehicles).

Start-up legislation and venture capital arrangements are also being pursued in Mali, Ghana, Rwanda, and many other (African Private Equity and Venture Capital Association, 2020). These tools are intended to support innovations for the transformation of the economy into a digital one. Through venture capital supports, firms like InstaDeep are delivering cutting edge artificial intelligence at the international level. The company operates in Nigeria, Tunisia and South Africa, employing hundreds of youths.

It is especially vital today and in the future to enhance employment opportunities in Africa, given the vast numbers of young people who are planning to join the workforce. The paper discovered that the 4IR technology allows for lower production costs in the economy, increased productivity and earnings, and the development and introduction of new business lines, resulting in many new opportunities that should be particularly appealing and accessible to Africa's youth. 4IR deployment of technology might produce new, typically formal, pay jobs, particularly in the service sector, which might be higher than the labor force increase and the salaries of the informal sector.

To get the most out of the 4IR, countries will have to digitalize their economy. The 4IR presents an opportunity for the informal-dominated African economies to turn to new business models that may not require the substantial capital provided by traditional enterprises.

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1. All stakeholders, including businesses, governments and industry regulators, must formulate and implement policies to assist everyone in developing the skills that technology complements.
 2. TVET reforms that are responsive to the market and demand-driven are required as components of a more effective economic strategy with well-defined implementation plans that include periodic evaluations.
 3. African countries must create a more integrated digital market throughout the continent to increase competition and possibilities for new and advanced startups.
 4. Young people in the workplace will need to take a life-long learning attitude to their jobs, updating their abilities regularly, either to stay on the cutting edge of their profession or to keep up with technological advancements across industries.
 5. Policymakers will need to perform an assessment of jobs in danger of being automated that considers the "Covid-19 New Normal." Workers will have access to opportunities in the post-pandemic digital economy if they have access to skill development and labour market information.

- Abe, E. N., Abe, I. I., & Adisa, O. (2021). Future of Work: Skill Obsolescence, Acquisition of New Skills, and Upskilling in the 4IR. In Future of Work, Work-Family Satisfaction, and Employee Well-Being in the Fourth Industrial Revolution, 217-231.
- Ayentimi, D. T., & Burgess, J. (2019). Is the fourth industrial revolution relevant to sub-Saharan Africa?. Technology analysis & strategic management, 31(6), 641-652.
- BBC. (2019). Will AI kill developing world growth? Retrieved from <https://www.bbc.com/news/business-47852589>.
- Brown, E., K. (2020). The Fourth Industrial Revolution: Will Africa be ready? African Center for Economic Transformation, Policy Briefs and Discussion Papers. Retrieved from [The Fourth Industrial Revolution: Will Africa be ready? - ACET \(acetforafrica.org\)](https://www.acetforafrica.org/)
- Cirera, X. (2016). Catching up to the technological frontier? Understanding firm-level innovation and productivity in Kenya. Innovation and Entrepreneurship, Trade and Competitiveness Global Practice, the World Bank, 94671, 1-45.
- Fal, M. (2013). Accelerating Entrepreneurship in Africa. Retrieved from <https://www.mitpressjournals.org> › pdf › INOV_a_00193
- Fififinance. (2020). Top 8 Startups in Kenya. <https://fififinance.com/africa/startups-kenya>.
- Fox, L., & Signé, L. (2021). The Fourth Industrial Revolution (4IR) and the Future of Work: Could this Bring Good Jobs to Africa?. Evidence Synthesis Paper Series.
- Frey, C., Osborne, M., & Holmes, C. (2016). Technology at Work v2.0: The Future is Not What it Used to Be, Citi GPS: Global Perspectives and Solutions. Oxford: University of Oxford, 338.
- Ibrahim Forum Report (2019). Africa's first challenge: the youth bulge stuck in 'waithood'. Retrieve from [Africa's first challenge: the youth bulge stuck in 'waithood' | Mo Ibrahim Foundation](https://www.ibrahimforum.org/).
- International Labour Organization. (2016). Facing the growing unemployment challenges in Africa. Retrieved from [https://www.ilo.org/africa/media-centre/pr/WCMS_444474/lang--en/index.htm](https://www.ilo.org/africa/media-centre/pr/WCMS_444474/lang-en/index.htm).
- Jules, T., D. (2017). The Global Educational Policy Environment in the Fourth Industrial Revolution: Gated, Regulated and Governed. Bingley: Emerald Group Publishing

- International Telecommunication Union. (2020). Digital trends in Africa 2021 Information and communication technology trends and developments in the Africa region 2017-2020. Retrieved from https://www.itu.int/dms_pub/itu-d/opb/ind/D-IND-DIG_TRENDS_AFR.01-2021-PDF-E.pdf
- Karr, J., Loh, K., & San Andres, E. (2020). COVID-19, 4IR and the future of work. APEC Policy Support Unit Brief, (34).
- Kudyba, S. (2020). COVID-19 and the Acceleration of Digital Transformation and the Future of Work. Information Systems Management, 37(4), 284-287.
- Makridakis, S. (2017). The Forthcoming Artificial Intelligence (AI) Revolution: Its Impact on Society and Firms. Futures 90, 46-60.
- Mckinsey Global Institute. (2017). Jobs lost, jobs gained: Workforce transitions in a time of automation. Report. Retrieved from [mgi jobs lost-jobs gained_report_december 2017.pdf](https://www.mckinsey.com/~/media/mckinsey/industries/automotive/automation/mgi-jobs-lost-jobs-gained-report-december-2017.pdf) ([mckinsey.com](https://www.mckinsey.com))
- Millington, K., A. (2017). How Changes in Technology and Automation Will Affect the Labour Market in Africa. Helpdesk report on knowledge, evidence and learning for development. Helpdesk Report. Retrieved
- Naudé, W. (2017). Entrepreneurship, education and the fourth industrial revolution in Africa. IZA Discussion Papers, No. 10855.
- Osei-Boateng, C., and E. Ampratwum. 2011. The Informal Sector in Ghana. Accra: Friedrich Ebert Stiftung-Ghana office.
- Pityana, S., M. (2019). How Africa can secure its long-term economic growth. World Economic Forum. Accessed from <https://www.weforum.org/agenda/2019/09/how-africa-can-secure-its-long-term-economic-growth/>
- Randow, J. (2016). Is the Outlook for Developing Nations Darkened by a Robot Future? Bloomberg. Retrieved from [Is the Outlook for Developing Nations Darkened by a Robot Future? - Bloomberg](https://www.bloomberg.com/news/articles/2016-09-27/is-the-outlook-for-developing-nations-darkened-by-a-robot-future)
- Schwab, K. (2016). The Fourth Industrial Revolution. Geneva: World Economic Forum.

Sutherland, E. (2020). The fourth industrial revolution—the case of South Africa. *Politikon*, 47(2), 233-252.

Techbuild Africa. (2019). Senegal Joins Tunisia in Passing the Startup Act. Accessed from

<https://techbuild.africa/senegal-joins-tunisia-in-passing-the-startup-act/>

The Economist. (2020). Africa's population will double by 2050. Special Report. Retrieve from

[Africa's population will double by 2050. The Economist.](#)

United Nations. (2017). World Population Prospects: The 2017 Revision. Retrieved from [World](#)

[Population Prospects: The 2017 Revision | Multimedia Library - United Nations Department of Economic and Social Affairs.](#)

World Economic Forum Africa. (2016). [World Economic Forum on Africa 2016. World Economic Forum \(weforum.org\)](#)

World Economic Forum. (2017). The Future of Jobs and Skills in Africa Preparing the Region for the Fourth Industrial Revolution. Doha: World Economic Forum.

www3.weforum.org/doc/WEF_EGW_FOJ_Africa.pdf.

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